SQL Sales Analytical project

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Project type: SQL Sales Analytical project

Database :Classicmodels database

Tools Used 🛠️:

- MySQL
- MySQL Workbench
- GitHub

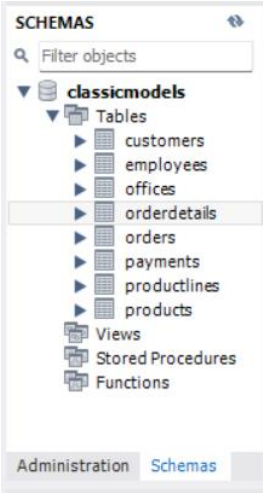
SQL Concepts Demonstrated 🎯:

- JOINs
- GROUP BY
- HAVING
- Aggregate Functions
- Subqueries
- Common Table Expressions(CTEs)
- Window Functions
- Ranking Functions

Project focus Area 📊:

- ✓ Customer Analysis
- ✓ Sales Analysis
- ✓ Product Analysis
- ✓ Revenue Analysis
- ✓ Employee Performance Analysis

Tables used:



---------CUSTOMER ANALYSIS--------

Question: LIST NUMBER OF CUSTOMERS IN EACH COUNTRY

Query:SELECT Country , count(customername) from customers group by country;

Output:

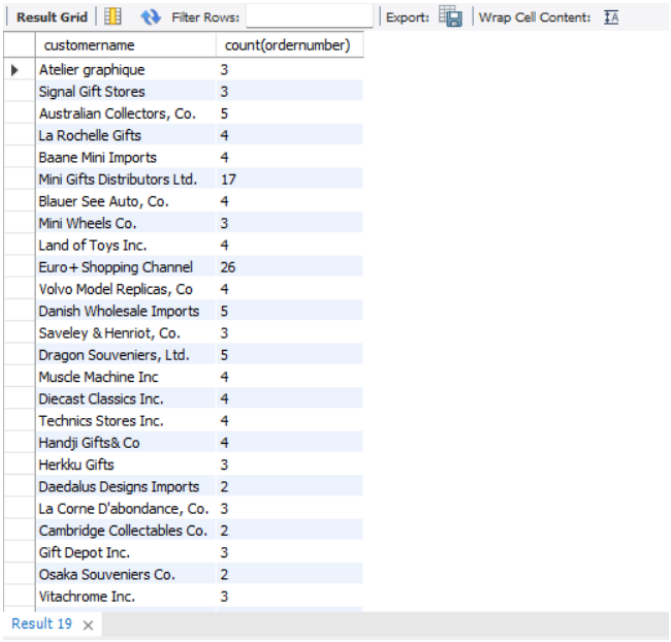
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	count(customername)
France	12
USA	36
Australia	5
Norway	1
Poland	1
Germany	13
Spain	7
Sweden	2
Denmark	2
Singapore	3
Norway	2
Portugal	2
Japan	2
Finland	3
UK	5
Ireland	2
Canada	3
Hong Kong	1
Italy	4
Switzerland	3
Netherla...	1
Belgium	2
New Zeal...	4
South Af...	1
Austria	2

Question: FIND NUMBER OF ORDERS PLACED BY EACH CUSTOMER

Query: select customername, count(ordernumber) from customers join orders using(customernumber) group by customername;

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

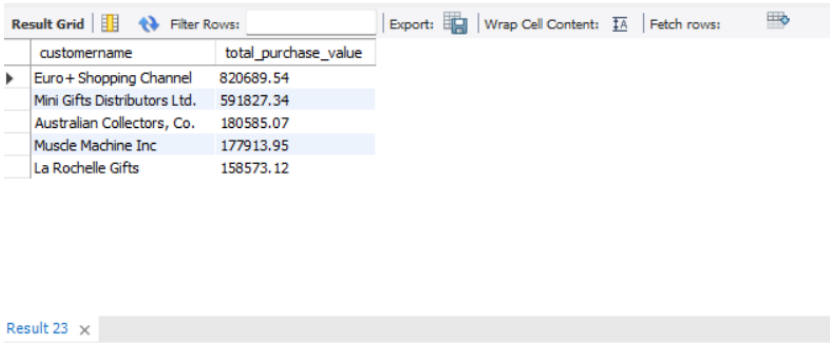
customername	count(ordernumber)
Atelier graphique	3
Signal Gift Stores	3
Australian Collectors, Co.	5
La Rochelle Gifts	4
Baane Mini Imports	4
Mini Gifts Distributors Ltd.	17
Blauer See Auto, Co.	4
Mini Wheels Co.	3
Land of Toys Inc.	4
Euro+ Shopping Channel	26
Volvo Model Replicas, Co	4
Danish Wholesale Imports	5
Saveley & Henriot, Co.	3
Dragon Souveniers, Ltd.	5
Muscle Machine Inc	4
Diecast Classics Inc.	4
Technics Stores Inc.	4
Handji Gifts& Co	4
Herkku Gifts	3
Daedalus Designs Imports	2
La Corne D'abondance, Co.	3
Cambridge Collectables Co.	2
Gift Depot Inc.	3
Osaka Souveniers Co.	2
Vitachrome Inc.	3

Result 19

Question: FIND TOP 5 CUSTOMERS BY TOTAL PURCHASE VALUE

Query: select customername , sum(priceeach*quantityordered) as total_purchase_value from customers join orders using(customernumber) join orderdetails using(ordernumber) group by customername;

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

customername	total_purchase_value
Euro+ Shopping Channel	820689.54
Mini Gifts Distributors Ltd.	591827.34
Australian Collectors, Co.	180585.07
Muscle Machine Inc	177913.95
La Rochelle Gifts	158573.12

Result 23

Question: CUSTOMER WHOSE TOTAL PURCHASE EXCEEDS CREDITLIMIT

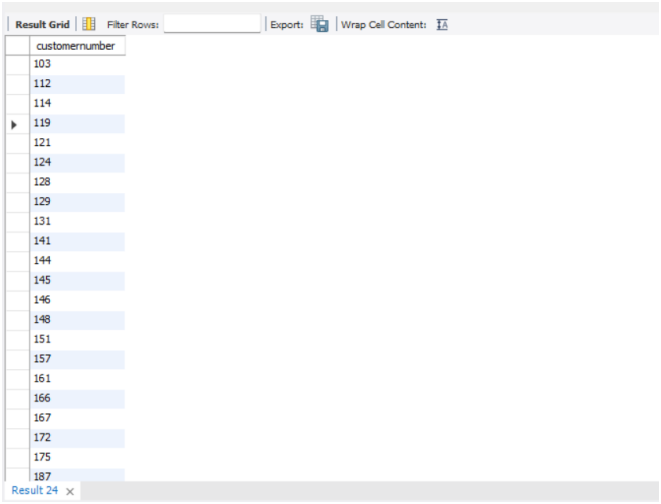
Query:

```
 with cte as

(select customernumber , sum(priceeach*quantityordered) as total_purchase
,creditlimit from customers join orders using(customernumber) join orderdetails
using(ordernumber) group by customerNumber)

select  customernumber from cte where total_purchase>creditLimit;
```

Output:



customernumber
103
112
114
119
121
124
128
129
131
141
144
145
146
148
151
157
161
166
167
172
175
187

Question: FIND CUSTOMERS WHO HAVE ORDERED PRODUCTS MORE THAN 3 PRODUCTLINES

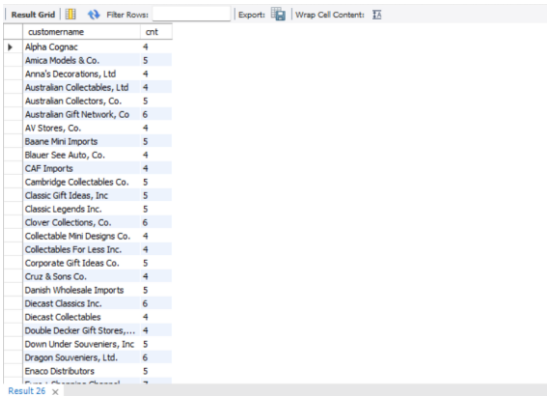
Query:

```
select customername,count(distinct(productline)) as cnt

from customers join orders using(customernumber) join orderdetails using(ordernumber) join
products using(productcode)

group by customername having cnt>3;
```

Output:



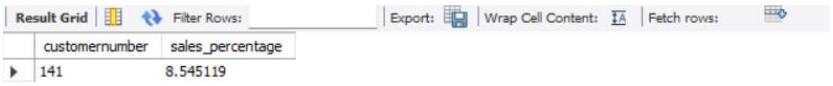
customername	cnt
Alpha Cognac	4
Amica Models & Co.	5
Anna's Decorations, Ltd	4
Australian Collectables, Ltd	4
Australian Collectors, Co.	5
Australian Gift Network, Co	6
AV Stores, Co.	4
Baane Mini Imports	5
Blauer See Auto, Co.	4
CAF Imports	4
Cambridge Collectables Co.	5
Classic Gift Ideas, Inc	5
Classic Legends Inc.	5
Clover Collections, Co.	6
Collectable Mini Designs Co.	4
Collectables For Less Inc.	4
Corporate Gift Ideas Co.	5
Cruz & Sons Co.	4
Danish Wholesale Imports	5
Diecast Classics Inc.	6
Diecast Collectables	4
Double Decker Gift Stores,...	4
Down Under Souveniers, Inc	5
Dragon Souveniers, Ltd.	6
Enaco Distributors	5

Question: FIND CUSTOMER CONTRIBUTING THE HIGHEST PERCENTAGE OF TOTAL SALES

Query:

```
select customernumber,
(sum(priceeach*quantityordered)*100/(select sum(priceeach*quantityordered)
from orderdetails)) as sales_percentage from customers join  orders using(customernumber)
join orderdetails using(ordernumber)
group by customernumber order by sales_percentage desc limit 1;
```

Output:



customernumber	sales_percentage
141	8.545119

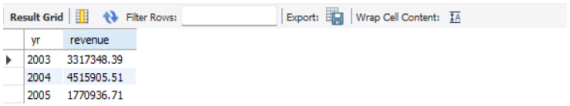
----------Sales & Revenue Analysis----------

Question: FIND TOTAL REVENUE GENERATED IN EACH YEAR

Query:

```
select year(orderdate) as yr, sum(priceeach*quantityordered) as revenue
from orders join orderdetails using(ordernumber) group by yr;
```

Output:



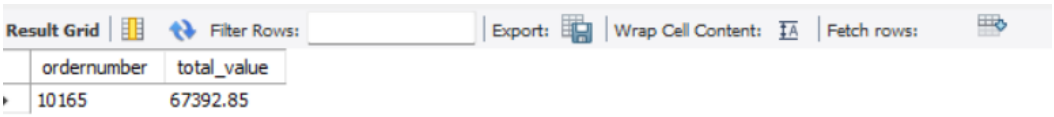
yr	revenue
2003	3317348.39
2004	4515905.51
2005	1770936.71

Question: SHOW ORDER WITH HIGHEST TOTAL VALUE

Query:

```
select ordernumber , sum(priceeach*quantityordered) as total_value
from orderdetails group by orderNumber order by total_value desc limit 1;
```

Output:



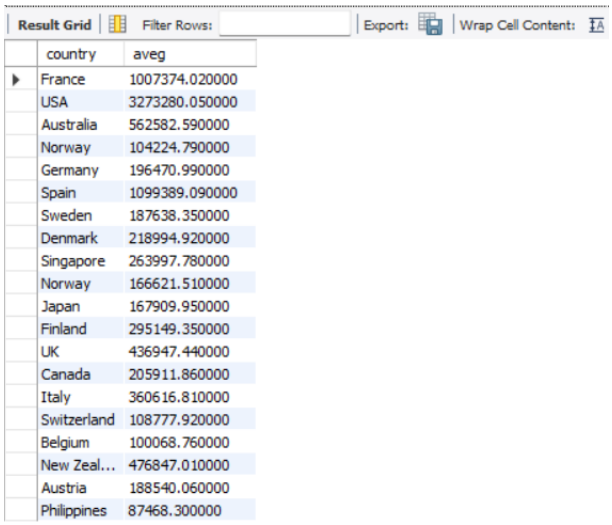
ordernumber	total_value
10165	67392.85

Question: DISPLAY COUNTRIES WHERE AVERAGE CUSTOMER PURCHASE EXCEEDS $50,000

Query:

```
(select country , sum(priceeach*quantityordered) as purchase
from customers join orders using(customernumber) join orderdetails using(ordernumber)
group by country)
select country , avg(purchase) as aveg from cte  group by country having aveg>50000;
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

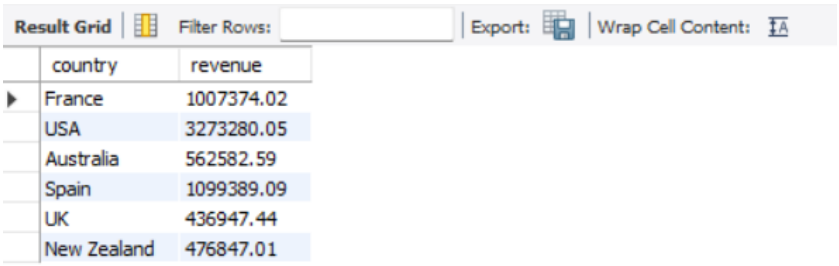
country	aveg
France	1007374.020000
USA	3273280.050000
Australia	562582.590000
Norway	104224.790000
Germany	196470.990000
Spain	1099389.090000
Sweden	187638.350000
Denmark	218994.920000
Singapore	263997.780000
Norway	166621.510000
Japan	167909.950000
Finland	295149.350000
UK	436947.440000
Canada	205911.860000
Italy	360616.810000
Switzerland	108777.920000
Belgium	100068.760000
New Zeal...	476847.010000
Austria	188540.060000
Philippines	87468.300000

Question: FIND COUNTRY WHOSE REVENUE IS ABOVE COMPARED TO OVERALL REVENUE

Query:

```
with cte as
(select country , sum(priceeach*quantityordered) as revenue
from customers join orders using(customernumber) join orderdetails
using(ordernumber) group by country)
select country, revenue  from  cte where revenue>(select avg(revenue) from cte);
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

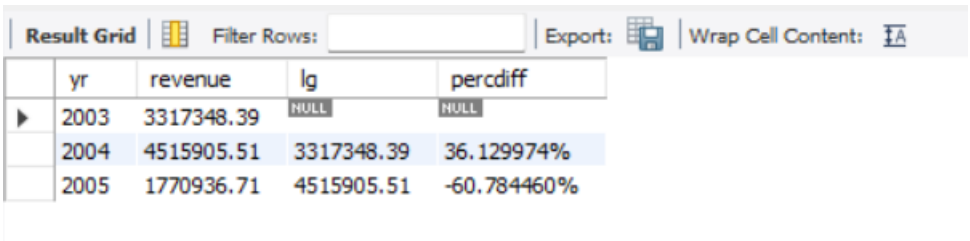
country	revenue
France	1007374.02
USA	3273280.05
Australia	562582.59
Spain	1099389.09
UK	436947.44
New Zealand	476847.01

Question: FIND TOTAL REVENUE GROWTH PERCENTAGE YEAR BY YEAR

Query:

```
with iie as
(With uue as
(select year(orderdate) as yr, sum(priceeach*quantityordered)as revenue
from customers join orders using(customernumber) join orderdetails using(ordernumber)
group by yr order by revenue desc)
select yr, revenue, lag(revenue) over (order by yr) as lg from uue)
select yr, revenue, lg, concat(((revenue-lg)/lg)*100,'%') as percdiff from iie;
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

yr	revenue	lg	percdiff
2003	3317348.39	NULL	NULL
2004	4515905.51	3317348.39	36.129974%
2005	1770936.71	4515905.51	-60.784460%

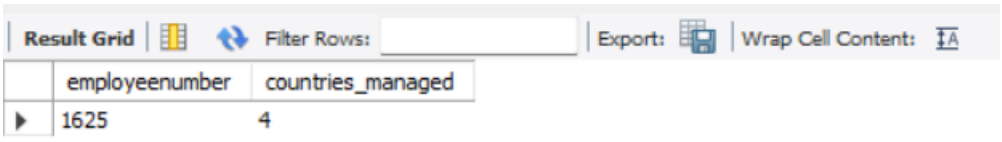
----------Employee Analysis----------

Question: FIND EMPLOYEES MANAGING CUSTOMERS FROM MULTIPLE COUNTRIES

Query:

```
select employeenumber ,  count(distinct(country)) as countries_managed
 from customers join employees on
customers.salesRepEmployeeNumber=employees.reportsto group by employeenumber
having countries_managed>1;
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeenumber	countries_managed
1625	4

Question: FIND EMPLOYEES AND NUMBER OF CUSTOMERS ASSIGNED TO THEM

Query:

```
select employeenumber, concat (firstname,' ', lastname) as
employee_name,count(customernumber) as cnt

from  employees left join customers c on employeenumber=c.salesRepEmployeeNumber

 group by employeenumber,employee_name order by cnt desc;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employeenumber	employee_name	cnt
1401	Pamela Castillo	10
1504	Barry Jones	9
1323	George Vanauf	8
1501	Larry Bott	8
1286	Foon Yue Tseng	7
1370	Gerard Hernandez	7
1165	Leslie Jennings	6
1166	Leslie Thompson	6
1188	Julie Firrelli	6
1216	Steve Patterson	6
1337	Loui Bondur	6
1702	Martin Gerard	6
1611	Andy Fixter	5
1612	Peter Marsh	5
1621	Mami Nishi	5
1002	Diane Murphy	0
1056	Mary Patterson	0
1076	Jeff Firrelli	0
1088	William Patterson	0
1102	Gerard Bondur	0
1143	Anthony Bow	0
1619	Tom King	0
1625	Yoshimi Kato	0

----------Product Analysis----------

Question: FIND PRODUCTLINE GENERATING HIGHEST REVENUE

Query:

```
select productline, sum(priceeach*quantityordered) as revenue

from products join orderdetails using(productcode) group by productline order by revenue
desc limit 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

productline	revenue
Classic Cars	3853922.49

Question: PRODUCTS THAT NEVER ORDERED

Query: select productname from products left join orderdetails using(productcode) left join orders using(ordernumber) where ordernumber is null;

Output:



productname
1985 Toyota Supra

Question: COUNT HOW MANY PRODUCTS BELONG TO PRODUCTLINE

Query: select productline , count(productname) from products group by productline;

Output:



productline	count(productname)
Classic Cars	38
Motorcycles	13
Planes	12
Ships	9
Trains	3
Trucks and Buses	11
Vintage Cars	24

Question: TOTAL SALES FOR EACH PRODUCTLINE

Query: select productline, sum(priceeach*quantity Ordered) as sale from orderdetails join products using(productcode) group by productline;

Output:



productline	sale
Classic Cars	3853922.49
Motorcycles	1121426.12
Planes	954637.54
Ships	663998.34
Trains	188532.92
Trucks and Buses	1024113.57
Vintage Cars	1797559.63

----------Advanced Analytics----------

Question: RANK CUSTOMERS BY TOTAL PURCHASE VALUE

Query:

```
with cte as
(select customername, sum(priceeach*quantityordered) as revenue
 from customers join orders using(customernumber) join orderdetails using(ordernumber)
 group by customername)
 select customername , revenue , rank() over ( order by revenue desc) as rnk from cte;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customername	revenue	rnk
Euro+ Shopping Channel	820689.54	1
Mini Gifts Distributors Ltd.	591827.34	2
Australian Collectors, Co.	180585.07	3
Muscle Machine Inc	177913.95	4
La Rochelle Gifts	158573.12	5
Dragon Souveniers, Ltd.	156251.03	6
Down Under Souveniers, Inc	154622.08	7
Land of Toys Inc.	149085.15	8
AV Stores, Co.	148410.09	9
The Sharp Gifts Warehouse	143536.27	10
Salzburg Collectables	137480.07	11
Kelly's Gift Shop	137460.79	12
Anna's Decorations, Ltd	137034.22	13
Souveniers And Things Co.	133907.12	14
Corporate Gift Ideas Co.	132340.78	15
Saveley & Henriot, Co.	130305.35	16
Danish Wholesale Imports	129085.12	17
Rovelli Gifts	127529.69	18
Reims Collectables	126983.19	19
L'ordine Souveniers	125505.57	20
Scandinavian Gift Ideas	120943.53	21
Online Diecast Creations Co.	116449.29	22
Corrida Auto Replicas, Ltd	112440.09	23
Vida Sport, Ltd	108777.92	24
Handji Gifts& Co	107746.75	25

Question: TOP 3 CUSTOMERS IN EACH COUNTRY

Query:

```
with mme as
(with gge as
(select  country ,customername , sum(priceeach*quantityordered) as revenue
from customers join orders using(customernumber) join orderdetails using(ordernumber)
group by country , customername order by revenue desc)
select country ,customername , revenue , rank() over (partition by country order by revenue desc ) as rnk from gge )
select * from mme where rnk<=3;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

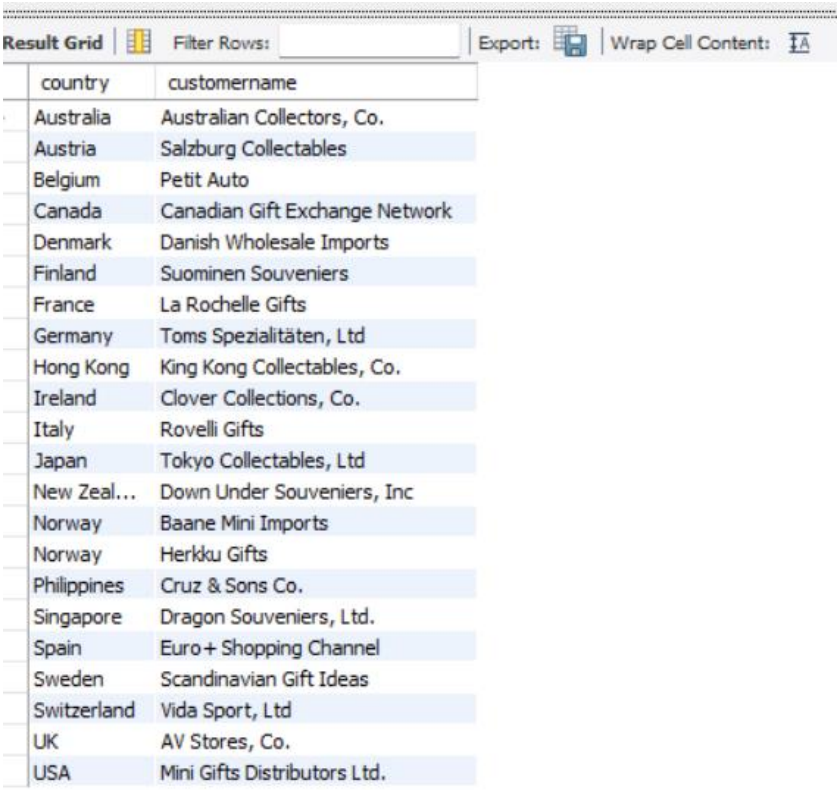
country	customername	revenue	rnk
Australia	Australian Collectors, Co.	180585.07	1
Australia	Anna's Decorations, Ltd	137034.22	2
Australia	Souveniers And Things Co.	133907.12	3
Austria	Salzburg Collectables	137480.07	1
Austria	Mini Auto Werke	51059.99	2
Belgium	Petit Auto	70851.58	1
Belgium	Royale Belge	29217.18	2
Canada	Canadian Gift Exchange Network	70122.19	1
Canada	Québec Home Shopping Network	68977.67	2
Canada	Royal Canadian Collectables, Ltd.	66812.00	3
Denmark	Danish Wholesale Imports	129085.12	1
Denmark	Heintze Collectables	89909.80	2
Finland	Suominen Souveniers	103896.74	1
Finland	Oulu Toy Supplies, Inc.	95706.15	2
Finland	Toys of Finland, Co.	95546.46	3
France	La Rochelle Gifts	158573.12	1
France	Saveley & Henriot, Co.	130305.35	2
France	Reims Collectables	126983.19	3
Germany	Toms Spezialitäten, Ltd	89223.14	1
Germany	Blauer See Auto, Co.	75937.76	2
Germany	Bavarian Collectables Imports, Co.	31310.09	3
Hong K...	King Kong Collectables, Co.	45480.79	1
Ireland	Clover Collections, Co.	49898.27	1
Italy	Rovelli Gifts	127529.69	1
Italy	L'ordine Souveniers	125505.57	2

Question: MOST VALUABLE CUSTOMER IN EACH COUNTRY

Query:

```
with mme as
(With cte as
(select country, customername , sum(priceeach*quantityordered)as revenue
from customers join orders using(customernumber) join orderdetails using(ordernumber)
group by country , customername )
select country, customername, revenue, rank() over (partition by country order by revenue desc) as rnk from cte)
select country, customername from mme where rnk=1;
```

Output:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	customername
Australia	Australian Collectors, Co.
Austria	Salzburg Collectables
Belgium	Petit Auto
Canada	Canadian Gift Exchange Network
Denmark	Danish Wholesale Imports
Finland	Suominen Souveniers
France	La Rochelle Gifts
Germany	Toms Spezialitäten, Ltd
Hong Kong	King Kong Collectables, Co.
Ireland	Clover Collections, Co.
Italy	Rovelli Gifts
Japan	Tokyo Collectables, Ltd
New Zeal...	Down Under Souveniers, Inc
Norway	Baane Mini Imports
Norway	Herkku Gifts
Philippines	Cruz & Sons Co.
Singapore	Dragon Souveniers, Ltd.
Spain	Euro+ Shopping Channel
Sweden	Scandinavian Gift Ideas
Switzerland	Vida Sport, Ltd
UK	AV Stores, Co.
USA	Mini Gifts Distributors Ltd.

This project utilized SQL to analyze customer behavior, product performance, employee assignments, and revenue trends using the ClassicModels database. Through the application of JOINs, GROUP BY, HAVING, Subqueries, CTEs, and Window Functions, meaningful business insights were derived that can support data-driven decision-making. The project strengthened practical SQL skills while demonstrating how SQL can be used to solve real-world business problems.